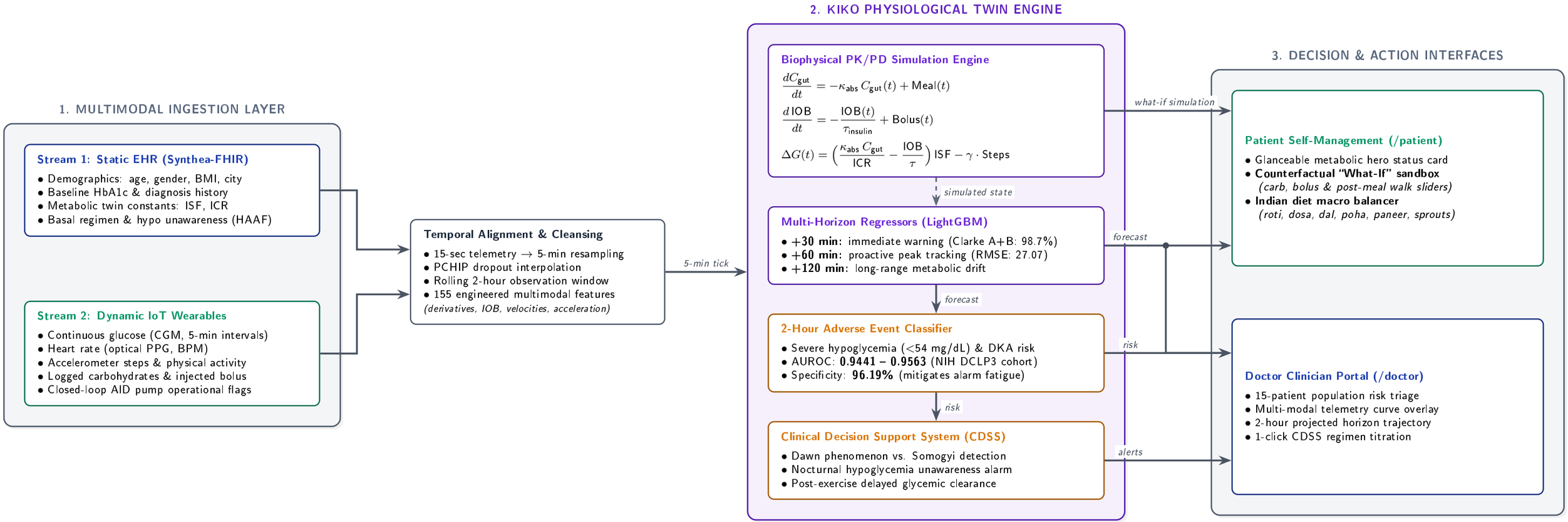

1. MULTIMODAL INGESTION LAYER
Stream 1: Static EHR (Synthea-FHIR)
• Demographics: age, gender, BMI, city
• Baseline HbA1c & diagnosis history
• Metabolic twin constants: ISF, ICR
• Basal regimen & hypo unawareness (HAAF)
Stream 2: Dynamic IoT Wearables
• Continuous glucose (CGM, 5-min intervals)
• Heart rate (optical PPG, BPM)
• Accelerometer steps & physical activity
• Logged carbohydrates & injected bolus
• Closed-loop AID pump operational flags
Temporal Alignment & Cleansing
• 15-sec telemetry → 5-min resampling
• PCHIP dropout interpolation
• Rolling 2-hour observation window
• 155 engineered multimodal features
(derivatives, IOB, velocities, acceleration)
5-min tick
2. KIKO PHYSIOLOGICAL TWIN ENGINE
Biophysical PK/PD Simulation Engine
$\frac{dC_{\text{gut}}}{dt} = -\kappa_{\text{abs}}\, C_{\text{gut}}(t) + \text{Meal}(t)$
$\frac{d\,\text{IOB}}{dt} = -\frac{\text{IOB}(t)}{\tau_{\text{insulin}}} + \text{Bolus}(t)$
$\Delta G(t) = \left(\frac{\kappa_{\text{abs}}\, C_{\text{gut}}}{\text{ICR}} - \frac{\text{IOB}}{\tau}\right)\text{ISF} - \gamma \cdot \text{Steps}$
simulated state
Multi-Horizon Regressors (LightGBM)
• +30 min: immediate warning (Clarke A+B: 98.7%)
• +60 min: proactive peak tracking (RMSE: 27.07)
• +120 min: long-range metabolic drift
forecast
2-Hour Adverse Event Classifier
• Severe hypoglycemia (<54 mg/dL) & DKA risk
• AUROC: 0.9441 – 0.9563 (NIH DCLP3 cohort)
• Specificity: 96.19% (mitigates alarm fatigue)
risk
Clinical Decision Support System (CDSS)
• Dawn phenomenon vs. Somogyi detection
• Nocturnal hypoglycemia unawareness alarm
• Post-exercise delayed glycemic clearance
3. DECISION & ACTION INTERFACES
what-if simulation
forecast
risk
alerts
Patient Self-Management (/patient)
• Glanceable metabolic hero status card
• Counterfactual "What-If" sandbox
(carb, bolus & post-meal walk sliders)
• Indian diet macro balancer
(roti, dosa, dal, poha, paneer, sprouts)
Doctor Clinician Portal (/doctor)
• 15-patient population risk triage
• Multi-modal telemetry curve overlay
• 2-hour projected horizon trajectory
• 1-click CDSS regimen titration